

SOFTWARE TEST PLAN (STP)

Electronics Store Inventory Management System (ESIMS)

Team: 10

Team Members: Vaishnavi Bandaru, Valmiki Uma, Yerramilli Siva Sai Saharsh, Yashas

SRN: PES2UG24CS570, PES2UG24CS573, PES2UG24CS618, PES2UG24CS609

1. Introduction

1.1 Purpose

This Software Test Plan defines the objectives, scope, strategy, resources, schedule, responsibilities, deliverables, risks, traceability and reporting for testing the Electronics Store Inventory Management System (ESIMS) version 1.0. It is derived from the ESIMS Software Requirements Specification (SRS) v1.0.

1.2 Scope

Testing covers authentication, product and inventory management, supplier management, purchases, automatic stock updates, sales, sale-total calculation, low-stock detection, product search, inventory and sales reports, performance, reliability, availability, usability and security. Customer-facing e-commerce, online payments, delivery management, advanced accounting, external e-commerce integration and barcode hardware are outside the current scope.

1.3 References

- ESIMS SRS v1.0, dated 06-09-2026.
- ESIMS Software Architecture and Design Specification v1.0, dated 14-09-2026.
- SRS Section 8 Requirements Traceability Matrix (RTM).
- Software Test Plan template supplied for the Software Engineering laboratory.

1.4 Definitions

Term	Definition
ESIMS	Electronics Store Inventory Management System
SKU	Stock Keeping Unit; unique product identifier
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix
RBAC	Role-Based Access Control
UAT	User Acceptance Testing

TC	
	Test Case

2. Test Items

- Authentication and Access Control
- Product Management
- Inventory Management
- Supplier Management
- Purchase Transactions
- Sales Transactions
- Reporting
- User Interface
- Database/transaction management
- Security and input validation

3. Features to be Tested

Requirement	Feature	Expected Result	Test Case
ESIMS-F-001			
	Login		
		Valid credentials allow authorized login.	
			TC-Auth-01
ESIMS-F-002			
	Invalid login		
		Invalid credentials are rejected.	
			TC-Auth-02
ESIMS-F-003			
	Add product		
		Valid product with name/category/price/quantity is added.	
			TC-Prod-01
ESIMS-F-004			
	Update product		
		Existing product information is updated correctly.	
			TC-Prod-02
ESIMS-F-005			
	View stock		
		Current stock quantity is displayed correctly.	
			TC-Inv-01
ESIMS-F-006			
	Search		
		Products can be found by name or SKU.	
			TC-Inv-02
ESIMS-F-007			
	Low stock		
		Products at/below reorder	

		level are identified.	
			TC-Inv-03
ESIMS-F-008			
	No negative stock		
		Sale/transaction exceeding stock is rejected.	
			TC-Inv-04
ESIMS-F-009			
	Suppliers		
		Supplier records can be added/updated.	
			TC-Sup-01
ESIMS-F-010			
	Purchase		
		Purchase from supplier is recorded.	
			TC-Pur-01
ESIMS-F-011			
	Purchase stock update		
		Stock increases by purchased quantity.	
			TC-Pur-02
ESIMS-F-012			
	Sale		
		Sale containing one or more products is recorded.	
			TC-Sale-01
ESIMS-F-013			
	Sale total		
		Total is correctly calculated from prices and quantities.	
			TC-Sale-02
ESIMS-F-014			
	Sale stock update		
		Stock decreases by quantity sold.	
			TC-Sale-03
ESIMS-F-015			
	Inventory report		
		Report shows products and current stock.	
			TC-Report-01
ESIMS-F-016			
	Sales report		
		Report shows sales for selected date/range.	
			TC-Report-02
ESIMS-NF-001			
	Performance		
		Common search/inventory operations meet 3-second target.	
			TC-Perf-01
ESIMS-NF-002			
	Reliability		
		Inventory remains consistent after transactions.	
			TC-Rel-01

ESIMS-NF-003			
	Availability		
		System remains accessible except planned maintenance.	
			TC-Avail-01
ESIMS-NF-004			
	Usability		
		Common tasks complete without critical usability errors.	
			TC-UX-01
ESIMS-NF-005			
	Data protection		
		Unauthorized access/modification is rejected.	
			TC-Sec-01
ESIMS-SR-001			
	Authentication security		
		Protected functions require authentication.	
			TC-Sec-02
ESIMS-SR-002			
	Role restriction		
		Restricted admin functions are denied to unauthorized roles.	
			TC-Sec-03
ESIMS-SR-003			
	Password hashing		
		Passwords are not stored as plaintext.	
			TC-Sec-04
ESIMS-SR-004			
	Input validation		
		Invalid input is safely rejected.	
			TC-Sec-05
ESIMS-SR-005			
	Transaction protection		
		Unauthorized record modification/deletion is prevented.	
			TC-Sec-06

4. Features Not to be Tested

- Customer-facing shopping/e-commerce interface.
- Online payment processing/payment gateways.
- Delivery or logistics management.
- Advanced accounting.
- External e-commerce integrations.
- Barcode scanner hardware, as it is a future enhancement.

5. Test Approach / Strategy

5.1 Test Levels

Level	Purpose	Example
Unit		
	Verify individual modules.	
		Authentication, validation, stock calculation.
Integration		
	Verify module interactions.	
		Purchase → stock update; sale → stock decrease.
System		
	Verify complete workflows.	
		Login → search → sale → report.
Acceptance/UAT		
	Verify SRS acceptance criteria.	
		High-priority requirements and major workflows.

5.2 Test Types

- Functional
- Integration
- System/end-to-end
- Regression
- Performance
- Reliability/transaction consistency
- Usability
- Security
- Authorization/role testing
- Negative and input-validation testing

5.3 Entry Criteria

- Testable build is available.
- Required modules are implemented.
- Dummy test data and test database are ready.
- Test environment is accessible.
- Test cases and expected results are reviewed.

5.4 Exit Criteria

- 100% of planned test cases executed.
- All high-priority requirements have test evidence.
- No Critical defects remain open.
- Inventory is accurate after tested purchases and sales.
- Security/authorization tests pass.
- Major workflows operate successfully.
- RTM is complete.

5.5 Security Validation

- Verify authentication before protected operations.

- Verify server-side role checks.
- Verify passwords are securely hashed, not plaintext.
- Test malformed/invalid input.
- Attempt unauthorized record modification/deletion.
- Verify protected transaction information is accessible only to authorized personnel.

6. Test Environment

Category	Environment
Hardware	
	Desktop/laptop, keyboard, mouse/touchpad, monitor.
Client	
	Modern web browser or supported desktop interface.
Application	
	ESIMS application build under test.
Database	
	Relational database used by ESIMS.
Network	
	Local/network environment as required.
Test Data	
	Dummy users, products, SKUs, suppliers, purchases, sales and stock.
Defect Tracking	
	Jira or equivalent, if execution tracking is required.
Optional Tools	
	Selenium, Postman and JMeter where applicable. Manual execution is acceptable unless automation is specifically required.

7. Test Schedule

Activity	Planned Date
Test planning/review	
	15-09-2026
Test case preparation	
	16-09-2026 to 17-09-2026
Environment/data setup	
	18-09-2026
Functional testing	
	19-09-2026 to 22-09-2026
Integration/system testing	
	23-09-2026 to 24-09-2026
Security/performance/usability	
	25-09-2026
Retesting/regression	
	26-09-2026 to 27-09-2026
Final summary/sign-off	
	28-09-2026

8. Test Deliverables

- Software Test Plan

- Test case document/sheet
- Test data
- Test execution results/evidence
- Defect reports, if applicable
- Updated RTM
- Final Test Summary Report

9. Roles and Responsibilities

Role	Responsibility
QA/Test Lead	
	Prepare plan, coordinate execution, review results and summary.
Test Engineer	
	Design/execute cases, capture evidence and report defects.
Developer	
	Fix defects and support test environment.
Business Representative	
	Validate store workflows and acceptance.
Project Team	
	Maintain requirements, design, implementation and RTM consistency.

10. Risks and Mitigation

Risk	Impact	Mitigation
Unstable build		
	Testing blocked.	
		Smoke test first; use stable build.
Incorrect stock update		
	Unreliable inventory.	
		Verify before/after every purchase and sale.
Concurrent last-unit sales		
	Overselling/negative stock.	
		Test concurrency and database locking.
Database failure		
	Operations unavailable.	
		Backups and restore procedure.
UI-only role checks		
	Privilege escalation.	
		Verify authorization server-side.
Large datasets		
	Response >3 seconds.	
		Representative data, indexes and bounded reports.
Limited time		
	Incomplete coverage.	
		Prioritize high-priority requirements.

11. Assumptions & Dependencies

- SRS and architecture/design specification are available.
- A testable build is supplied before execution.
- Relational test database is available.
- Only dummy test data is used.
- Both Store Manager/Administrator and Store Staff/Sales Staff roles are available for testing.
- Barcode hardware is not required for the current scope.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable for a prolonged period.
- Blocking defects prevent execution of about 30% or more of planned cases.
- Database/test data is unavailable or corrupted.

Resume testing when:

- Blocking defects are fixed or a stable build is supplied.
- Environment/database is restored.
- Required test data is verified.

13. Test Case Management & Traceability

Each SRS requirement is mapped to one or more test cases. Execution records should include preconditions, data, steps, expected result, actual result, status and defect ID where applicable.

Test Case	Requirement	Objective	Input/Condition	Expected Result
TC-Auth-01				
	ESIMS-F-001			
		Valid login		
			Valid credentials	
				Dashboard opens.
TC-Auth-02				
	ESIMS-F-002			
		Invalid login		
			Wrong credentials	
				Login rejected with error.
TC-Prod-01				
	ESIMS-F-003			
		Add product		
			Unique SKU + valid details	
				Product saved.
TC-Prod-02				
	ESIMS-F-004			
		Update product		
			Existing product	
				Changes persist.
TC-Inv-01				
	ESIMS-F-005			
		View stock		
			Existing product	

				Correct quantity displayed.
TC-Inv-02				
	ESIMS-F-006			
		Search		
			Name/SKU	
				Matching products displayed.
TC-Inv-03				
	ESIMS-F-007			
		Low stock		
			Quantity $\leq$ reorder level	
				Product flagged.
TC-Inv-04				
	ESIMS-F-008			
		Prevent negative stock		
			Sale > available stock	
				Transaction rejected; stock unchanged.
TC-Sup-01				
	ESIMS-F-009			
		Manage supplier		
			Valid supplier	
				Added/updated successfully.
TC-Pur-01				
	ESIMS-F-010			
		Record purchase		
			Valid purchase	
				Purchase recorded.
TC-Pur-02				
	ESIMS-F-011			
		Purchase stock update		
			Known stock + purchase	
				Stock increases exactly.
TC-Sale-01				
	ESIMS-F-012			
		Record sale		
			Available product	
				Sale recorded.
TC-Sale-02				
	ESIMS-F-013			
		Calculate total		
			Known prices/quantities	
				Expected total matches.
TC-Sale-03				
	ESIMS-F-014			
		Sale stock update		
			Known stock + sale	

				Stock decreases exactly.
TC-Report-01				
	ESIMS-F-015			
		Inventory report		
			Known products/stock	
				Current inventory shown.
TC-Report-02				
	ESIMS-F-016			
		Sales report		
			Date/date range	
				Correct sales shown.
TC-Perf-01				
	ESIMS-NF-001			
		Response time		
			Normal load	
				90% of common requests ≤ 3 sec.
TC-Rel-01				
	ESIMS-NF-002			
		Inventory consistency		
			Purchase/sale sequence	
				Final values correct; no partial update.
TC-Avail-01				
	ESIMS-NF-003			
		Availability		
			Normal environment	
				System accessible.
TC-UX-01				
	ESIMS-NF-004			
		Usability		
			Common tasks	
				No critical usability errors.
TC-Sec-01				
	ESIMS-NF-005			
		Data protection		
			Unauthorized attempt	
				Attempt rejected.
TC-Sec-02				
	ESIMS-SR-001			
		Authentication enforcement		
			Unauthenticated request	
				Protected function inaccessible.
TC-Sec-03				
	ESIMS-SR-002			

		Role restriction		
			Staff tries restricted admin action	
				Access denied.
TC-Sec-04				
	ESIMS-SR-003			
		Password hashing		
			Inspect credential storage	
				No plaintext password.
TC-Sec-05				
	ESIMS-SR-004			
		Input validation		
			Malformed input	
				Input safely rejected.
TC-Sec-06				
	ESIMS-SR-005			
		Transaction protection		
			Unauthorized update/delete	
				Operation rejected; data unchanged.

13.1 RTM Summary

Requirement	Test Case	Status
ESIMS-F-001		
	TC-Auth-01	
		Not Yet Tested
ESIMS-F-002		
	TC-Auth-02	
		Not Yet Tested
ESIMS-F-003		
	TC-Prod-01	
		Not Yet Tested
ESIMS-F-004		
	TC-Prod-02	
		Not Yet Tested
ESIMS-F-005		
	TC-Inv-01	
		Not Yet Tested
ESIMS-F-006		
	TC-Inv-02	
		Not Yet Tested
ESIMS-F-007		
	TC-Inv-03	
		Not Yet Tested
ESIMS-F-008		
	TC-Inv-04	
		Not Yet Tested
ESIMS-F-009		
	TC-Sup-01	
		Not Yet Tested
ESIMS-F-010		

	TC-Pur-01	
		Not Yet Tested
ESIMS-F-011		
	TC-Pur-02	
		Not Yet Tested
ESIMS-F-012		
	TC-Sale-01	
		Not Yet Tested
ESIMS-F-013		
	TC-Sale-02	
		Not Yet Tested
ESIMS-F-014		
	TC-Sale-03	
		Not Yet Tested
ESIMS-F-015		
	TC-Report-01	
		Not Yet Tested
ESIMS-F-016		
	TC-Report-02	
		Not Yet Tested
ESIMS-NF-001		
	TC-Perf-01	
		Not Yet Tested
ESIMS-NF-002		
	TC-Rel-01	
		Not Yet Tested
ESIMS-NF-003		
	TC-Avail-01	
		Not Yet Tested
ESIMS-NF-004		
	TC-UX-01	
		Not Yet Tested
ESIMS-NF-005		
	TC-Sec-01	
		Not Yet Tested
ESIMS-SR-001		
	TC-Sec-02	
		Not Yet Tested
ESIMS-SR-002		
	TC-Sec-03	
		Not Yet Tested
ESIMS-SR-003		
	TC-Sec-04	
		Not Yet Tested
ESIMS-SR-004		
	TC-Sec-05	
		Not Yet Tested
ESIMS-SR-005		
	TC-Sec-06	
		Not Yet Tested

14. Test Metrics & Reporting

Metric	Formula/Purpose
Execution %	

	Executed / planned × 100.
Pass %	
	Passed / executed × 100.
Fail %	
	Failed / executed × 100.
Requirement coverage	
	Requirements with executed test cases.
Defect severity	
	Count Critical/High/Medium/Low defects.
Defect aging	
	Time defects remain open.
Performance compliance	
	% measurements meeting 3-second target.

14.1 Reports

- Periodic execution status
- Defect status report
- Updated RTM
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA/Test Lead		
Developer/Dev Lead		
Business Representative		
Course Coordinator		

End of Software Test Plan